

Semester 2026 AI Integration Playbook

Purpose: Map agentic workflows, governance, vendor evaluation, institutional data safeguards, ROI measurement, and student onboarding for Semester—the connected education operating system.

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Scope: This is a build and operating playbook for a higher-education platform. It is not legal advice, a compliance certification, or permission to process institutional education records without an appropriate agreement and institutional authorization.

1. Executive model

Semester Intelligence should be a **source-aware, policy-aware, permission-aware orchestration layer** across the product—not a general chatbot and not an autonomous decision maker.

```
Student / staff request
  → identity, tenant, role, and consent check
  → workflow policy check
  → minimum necessary context selection
  → source retrieval and source-access verification
  → model/provider routing
  → structured output validation
  → citations, uncertainty, and limitation checks
  → preview and confirmation for consequential actions
  → audit event, feedback, monitoring, and optional memory update
```

Non-negotiable AI rules

1. AI never decides authorization or bypasses database policy.
2. AI receives only the minimum data required for the current task.
3. AI may explain, draft, and prepare; it may not silently send, share, schedule, enroll, pay, delete, or change records.
4. Every consequential action requires an exact preview and explicit user confirmation.
5. Institutional data is processed only under the institution's approved purpose, agreement, data scope, policy, and retention rule.
6. AI outputs based on institutional or course materials must show source anchors or disclose missing sources.
7. Uploaded material is treated as untrusted data—not executable instruction.
8. AI must respect academic-integrity policy at institution, course, and assignment levels.
9. Students can see, correct, delete, and control eligible AI history and memory.

10. Every agentic workflow requires an owner, evaluation set, audit events, abuse controls, kill switch, and rollback plan.

2. Agentic workflow framework

An agentic workflow is not “give an AI access to everything.” It is a narrow, constrained sequence with a defined job, explicit inputs, specific tools, approval gates, and evidence.

Standard workflow card

```
Workflow name
Target user and job
Trigger
Allowed inputs
Allowed source systems
Data classification allowed
Tools permitted
Tools prohibited
Output schema
Citation requirement
Policy and integrity checks
Human approval requirements
Audit events
Success metric
Failure fallback
Kill switch
Owner
```

Risk tiers

Tier	Description	Examples	Required control
A	Read-only explanation	Summarize a selected syllabus	Source check and output validation
B	Draft creation	Draft an advisor agenda or study plan	User review before saving/sharing
C	Internal state preparation	Propose Action Center task or plan draft	Explicit user approval before persistence
D	External/reversible action	Create calendar draft, export a document	Exact preview and confirmation
E	High-impact or regulated action	Send message, share records, payment handoff, enrollment action	Fresh auth, exact confirmation, audit, often official handoff only

3. Ten agentic workflows

These workflows span product operations, engineering, institutional customer success, sales, and internal hiring/people operations. They are ordered to maximize value while limiting risk.

Workflow 1 — Student Path and Registration Copilot

Field	Design
User	Student
Job	Understand requirements, compare courses, prepare registration
Trigger	Student opens My Path, asks a question, or registration window approaches
Inputs	Student-entered goals, approved program requirements, selected courses, schedule constraints
Sources	Verified catalog/SIS read data, student plan, course schedule, official registration window
Allowed actions	Explain, compare, create a plan draft, propose backups, draft advisor questions
Prohibited actions	Certify degree completion, guarantee eligibility/seat availability, register/add/drop/withdraw
Approval	Student confirms saved plan; official registration is a handoff
ROI measures	Plan completion, backup-course rate, conflict-resolution rate, advisor-agenda rate, clarity score
Key safeguards	Estimate labels, source freshness, clear limitations, official fallback

Workflow 2 — Syllabus-to-Study Studio Agent

Field	Design
User	Student, faculty-approved course context
Job	Transform authorized course materials into study assets
Trigger	Student uploads/selects syllabus, slides, readings, notes, audio, or video
Inputs	Selected materials and course policy
Sources	Authorized course materials with stable page/slide/timestamp anchors
Allowed actions	Extract deadlines for review, generate summaries, notes, flashcards, quizzes, study plans, explainers
Prohibited actions	Fabricate sources, process disallowed material, answer active prohibited assessment questions
Approval	Student confirms extracted deadlines/actions; generated assets are editable
ROI measures	Study asset creation, study completion, source-open rate, helpfulness, citation coverage
Key safeguards	Malware/type scan, OCR/extraction review, course AI policy, source citation, prompt-injection defense

Workflow 3 — Academic Integrity Guardrail Agent

Field	Design
User	Student, faculty, institution
Job	Provide allowed learning help while preventing prohibited assessment assistance
Trigger	Student asks for support in a course/assignment context
Inputs	Course policy, assignment metadata when authorized, request content, sources
Sources	Faculty/institution policy, approved course context

Field	Design
Allowed actions	Explain concepts, create analogous practice, interpret rubric, build study plan, clarify feedback
Prohibited actions	Complete active graded submissions, fabricate sources/data, assist plagiarism evasion
Approval	No external action; user can request explanation or safe alternative
ROI measures	Safe-alternative completion, integrity-policy compliance, faculty trust, false-block rate
Key safeguards	Policy hierarchy, human escalation, feedback for incorrect classification, auditable refusals

Workflow 4 — Advisor Meeting Preparation Agent

Field	Design
User	Student; advisor sees only student-approved share
Job	Prepare a focused advising meeting and follow-up workflow
Trigger	Student schedules/creates an advisor meeting
Inputs	Student-selected plan, questions, scenarios, deadlines, goals
Sources	Student-selected content; verified requirements when connected
Allowed actions	Draft agenda, summarize open decisions, propose questions, create follow-up drafts
Prohibited actions	Reveal private study, health, financial, or non-shared data to advisor
Approval	Student reviews agenda and scope before sharing; advisor follow-up requires student confirmation or separate consent
ROI measures	Agenda creation, advisor usefulness score, meeting preparedness, follow-up completion
Key safeguards	Granular sharing scope, expiry, revocation, view audit, no default record access

Workflow 5 — Campus Support Navigator Agent

Field	Design
User	Student
Job	Find the appropriate campus resource and prepare a safe handoff
Trigger	Student asks where to get help or describes a non-emergency issue
Inputs	Student question and selected preferences
Sources	Verified campus graph: offices, services, policies, events, contacts, hours, accessibility info
Allowed actions	Explain options, identify relevant office, draft questions, save resource, create action, open official link
Prohibited actions	Diagnosis, emergency triage promises, legal/medical/financial determinations, disclosure of restricted data
Approval	Student approves any saved action, scheduling, or sharing
ROI measures	Search-to-action conversion, successful handoffs, support discovery, student clarity
Key safeguards	Source freshness, emergency disclaimer/routing, scoped data, office ownership, stale-content alerts

Workflow 6 — Career Evidence and Opportunity Agent

Field	Design
User	Student
Job	Turn confirmed work into portfolio evidence and prepare career actions
Trigger	Student uploads/selects project, course work, experience, or job description
Inputs	Student-confirmed artifact, selected goals, job description, resume/portfolio data
Sources	Student-owned work, verified skills records, approved opportunity listings
Allowed actions	Suggest skills, draft bullet points, organize portfolio, compare job requirements, prepare interview questions
Prohibited actions	Invent achievements, apply automatically, rank students for employers, infer protected traits
Approval	Student accepts/edits/rejects all skill mappings and drafts; any employer visibility is separate opt-in
ROI measures	Portfolio artifacts saved, confirmed skills, application milestones, mentor meetings
Key safeguards	Student control, transparent match reasons, expiry for talent profiles, view logs

Workflow 7 — Institutional Content Governance Agent

Field	Design
User	Authorized campus content owners and administrators
Job	Keep campus resources, policies, events, and opportunities accurate and accessible
Trigger	New content, stale content, accessibility issue, expiring event, broken link
Inputs	Approved institution content and owner metadata
Sources	CMS, directory, event feeds, policy records, official URLs
Allowed actions	Detect missing fields, draft summaries, identify stale pages, create accessibility review checklist, prepare publication draft
Prohibited actions	Auto-publish policy changes, override content owner, alter official record without approval
Approval	Named content owner approves publication; console records version/audit event
ROI measures	Content freshness, broken-link reduction, publication time, accessibility completion
Key safeguards	Owner approval, version history, source URL, scheduled review, public/private classification

Workflow 8 — Engineering Reliability Triage Agent

Field	Design
User	Engineering/SRE/security team
Job	Summarize incidents, correlate errors, propose runbook steps, generate test cases
Trigger	Runtime-error cluster, failed deployment, integration failure, alert, support spike

Field	Design
Inputs	Sanitized logs, traces, deployment metadata, runbooks, issue tracker data
Sources	Vercel/Supabase logs, monitoring, GitHub issues, internal runbooks
Allowed actions	Summarize, cluster, propose remediation, draft incident update, suggest regression tests
Prohibited actions	Deploy code, alter production data, rotate credentials, change firewall, close incidents without human approval
Approval	Human engineer approves ticket, code change, deployment, or communication
ROI measures	Mean time to detect, mean time to mitigate, recurring-incident rate, test coverage added
Key safeguards	Log redaction, least-privilege read-only connectors, no secret exposure, audit trail

Workflow 9 — Institutional Sales and RFP Copilot

Field	Design
User	Sales, partnerships, implementation team
Job	Prepare accurate RFP answers, pilot scopes, security responses, implementation plans
Trigger	New RFP, discovery call, procurement questionnaire, pilot proposal
Inputs	Approved product facts, trust artifacts, pricing rules, integration catalog, customer-approved case studies
Sources	Versioned Trust Room, approved answer library, CRM data, contract templates
Allowed actions	Draft response, map requirements, identify evidence gaps, prepare discovery questions, produce proposal draft
Prohibited actions	Make unsupported compliance claims, quote unapproved pricing, sign contracts, disclose protected customer data
Approval	Sales/legal/security owner reviews all external submissions
ROI measures	RFP response cycle time, security-review cycle time, pilot conversion, proposal accuracy
Key safeguards	Claims library, document version controls, approval workflow, customer confidentiality controls

Workflow 10 — Talent and Hiring Operations Copilot

Field	Design
User	Internal recruiting/hiring managers
Job	Organize role requirements, interview logistics, onboarding material, and candidate communications
Trigger	Open role, interview loop, new hire, performance-development process
Inputs	Approved job description, interview rubric, candidate-provided materials, internal policies
Sources	ATS/HRIS only if approved; role templates; internal hiring documentation
Allowed actions	Draft interview kits, summarize candidate-provided materials, prepare onboarding checklist, draft communications

Field	Design
Prohibited actions	Automated hiring decisions, ranking based on protected characteristics, inferring sensitive traits, sending offers without approval
Approval	Human hiring manager approves all decisions and communications
ROI measures	Time-to-fill, interview preparation time, candidate experience, onboarding completion
Key safeguards	Human decision requirement, data minimization, retention rules, access controls, bias evaluation

4. Workflow mapping method

4.1 Start with the human job—not the model

For every proposed AI feature, map this sequence:

```
User role
→ real problem and trigger
→ current workaround
→ minimum useful AI assistance
→ source systems needed
→ permissions/consent
→ output and action risk
→ human approval
→ measurable result
→ safe fallback
```

4.2 Example: “Help me prepare for registration”

```
Trigger:
  Registration opens in 9 days.

Context selected:
  Student’s saved plan, target term, approved requirement/curriculum data,
  course catalog, schedule constraints, current registration window.

AI work:
  Explain open requirement; identify eligible courses; compare schedule impact;
  propose primary and backup choices; draft advisor questions.

Output:
  Reviewable registration-readiness board.

Student confirmation:
  Save course plan, save backups, create advisor agenda.

Official boundary:
  Semester opens official registration portal; it does not register the student.
```

Audit:

Sources used, policy outcome, output version, actions approved, consent scope.

4.3 Workflow implementation sequence

1. Define the job, trigger, and non-AI baseline.
2. Write the workflow card and risk-tier classification.
3. Define data classification and sources.
4. Enforce authorization before retrieval—not after generation.
5. Create typed input/output schemas.
6. Add source anchors and citation requirements.
7. Add policy and integrity checks.
8. Add action permissions and confirmation binding.
9. Build negative tests and adversarial prompt-injection tests.
10. Instrument quality, cost, latency, and user feedback.
11. Ship behind tenant/role/cohort feature flags.
12. Review evidence before broad rollout.

5. Institutional data and privacy design

5.1 Data categories

Classification	Examples	AI handling rule
Public	Published catalog, public event, public policy	May be used in approved retrieval with source citation
Internal	Institution operations documentation	Use only with tenant authorization and need-to-know access
Student private	Student plan, goals, drafts, study activity, portfolio	Use only for the student's selected task; never expose to staff by default
Restricted	Education records, financial details, accommodations, health, conduct, authentication secrets	Exclude by default; require narrow approved workflow, explicit authority, strict logs, and often no AI processing

5.2 Institutional data decision tree

Is this institutional data?

No → apply standard user privacy policy.

Yes → is there an approved institutional agreement and authorized purpose?

No → do not ingest/process it.

Yes → is the requested field minimum necessary?

No → reduce scope.

Yes → is it restricted/sensitive?

Yes → exclude from AI by default; require special policy, approval, and

No → does the user have role, tenant, object, and consent authorization
No → deny without revealing data.
Yes → retrieve only task-scoped data, label source/freshness, process

5.3 Institutional data safeguards

Institution-approved purpose limitation
Written agreement and direct institutional control where required
Tenant isolation
Field-level mapping and minimum necessary data
Role + capability + scope + object authorization
Student consent for student-controlled sharing
Explicit expiry and revocation
Encryption in transit and at rest
Server-side secrets and provider credentials
No public-model training on institutional/student data without explicit documented approval
Provider data-use verification
Source/freshness labels
Audit logs for sensitive access, prompts, actions, sharing, export, and deletion
Retention schedule and verified deletion
Data export and correction workflows
Incident response and breach notification process

5.4 AI provider requirements

Before an AI provider is approved, require:

Requirement	Verification
Contractual data-use restrictions	Legal/security review
No unauthorized training on customer data	Provider terms and configuration evidence
Encryption and access controls	Security documentation
Regional/data-residency suitability if required	Provider documentation
Retention controls	Provider configuration and agreement
Subprocessor disclosure	Vendor documentation
Incident notification obligations	Contract/legal review
Logging and audit support	Technical evaluation
Model/version change controls	Provider change process
Reliability and fallback	SLO evidence and routing plan
Cost controls	Rate limits, quotas, usage reporting
Prompt-injection/tool-use protections	Security test evidence

5.5 What must never be sent casually to AI

Do not include these in prompts by default:

- Raw passwords, secrets, API keys, session tokens, or credentials.
- Unnecessary full student records.
- Health, diagnosis, disability, conduct, disciplinary, or emergency details.
- Detailed financial-aid records, payment credentials, bank data, or account ledger detail.
- Private advisor notes or counseling information.
- Full class rosters or student data when aggregate or de-identified data will work.
- Employment hiring decisions or protected-trait data.
- Any information the user has not authorized for the active task.

6. AI vendor evaluation scorecard

Score each prospective AI vendor or model provider 0–5. Do not approve a provider based on model quality alone.

Dimension	Weight	What “5” means
Data-use restrictions	15%	Contractually no training/secondary use without explicit authorization
Privacy and retention controls	10%	Configurable retention, deletion, access controls, clear subprocessors
Security posture	10%	Strong documented security, encryption, auditability, incident process
Education-policy fit	8%	Supports source grounding, academic-integrity controls, policy routing
Source and citation support	8%	Structured retrieval, stable citations, provenance support
Tool-use safety	8%	Narrow tools, confirmation controls, guardrails against excessive agency
Prompt-injection resilience	8%	Tested mitigations and safe context/tool boundaries
Model quality	8%	Reliable performance on Semester evaluation tasks
Accessibility capability	5%	Supports captions, structured output, multilingual/plain-language use cases
Reliability and latency	5%	SLOs, fallbacks, regional availability, predictable performance
Cost transparency	5%	Clear pricing, quotas, monitoring, controllable unit economics
Portability	5%	Provider-agnostic architecture, exportable prompts/evals, low lock-in

Approval thresholds

Overall weighted score: 4.0 / 5 minimum for production use
Data-use restrictions: 4.0 / 5 minimum
Privacy and retention: 4.0 / 5 minimum
Security posture: 4.0 / 5 minimum

Tool-use safety: 4.0 / 5 minimum for agentic workflows
No critical legal/security/privacy blocker may be offset by high model quality

NIST / 1EdTech alignment

Framework theme	Semester implementation
Govern	AI inventory, owner, policy, risk register, vendor approval, review cadence
Map	Workflow cards, stakeholders, data classes, harms, source systems, use cases
Measure	Evaluations, citation accuracy, safety tests, bias/equity checks, cost/latency metrics
Manage	Feature flags, model routing, kill switches, incident response, remediation, user controls
Transparency	AI disclosures, source labels, provider register, limitation notices
Choice	Opt-in/out where appropriate, source selection, history deletion, consent controls
Accountability	Human approval for consequential action; audit logs; named owners

7. Student onboarding experience

7.1 Onboarding principle

Do not begin by asking the student to configure every feature. Begin by delivering a useful result in minutes.

First-session goal: “I now know what I need to do next.”

7.2 Five-minute onboarding flow

Step 1 — Welcome and trust

Welcome to Semester.
One connected place for your university life.

Before we begin:

- You control what you share.
- You can see where important information comes from.
- Semester will label estimates and ask before taking consequential actions.
- You can change privacy, accessibility, and AI settings anytime.

Actions:

- Create account or use institutional SSO.
- Verify email if applicable.
- Show concise Privacy, AI, Accessibility, and support links.
- Ask whether the student is using an institution-sponsored account or personal account.

Step 2 — Choose a starting point

What would be most helpful today?

[Plan my next term]
[Understand my academic path]
[Organize this week]
[Get help with a course]
[Find campus support]
[Explore career options]

This choice routes the student to the smallest relevant onboarding path. Do not force every student through the same sequence.

Step 3 — Build minimum context

Ask only for what is necessary:

School or institution
Program/major or area of interest
Current term or target term
Expected graduation term if known
Primary goal today
Optional: preferred study times / work or travel constraints

Display labels for information source:

Student entered
Institution verified
Imported
Estimated
Needs review

Step 4 — Deliver the first useful result

Examples:

Starting choice	First outcome
Plan my next term	Basic term plan and visible weekly schedule
Understand my path	Path Snapshot with what is known, estimated, and needs review
Organize this week	Today briefing and top three actions
Get help with a course	Course workspace or study plan using selected materials
Find campus support	Verified resource recommendation and safe handoff
Explore career	Starter goal, related skills, opportunity or portfolio checklist

Step 5 — Explain the result

Every recommendation should include:

```
Why this appeared
Data used
What is estimated or missing
What you can change
What to do next
```

Example:

```
Next best step: Choose a quantitative-reasoning course for Fall.

Why: Your saved plan has one open requirement and room for three credits.
Data used: Your target term, saved program, planned Fall courses.
Limitation: Course eligibility and availability must be confirmed by your institution

[ Explore options ] [ Ask Semester ] [ Not relevant ]
```

Step 6 — Invite, do not demand, connections

Only after initial value:

```
Connect your calendar
Select course materials
Invite an advisor to view a specific agenda
Join a study group
Find a campus event
Set accessibility preferences
Choose AI source and history controls
```

Every connection uses incremental consent, explains scope, shows disconnect/delete-imported-data controls, and avoids broad permissions.

7.3 AI onboarding

Introduce AI as an inspectable helper:

```
Ask Semester can help you understand, plan, study, prepare, and find support.

It will:
• Use sources you select or that your institution has approved.
• Show sources and limitations when they are available.
• Follow your course and institution AI rules.
• Ask before it saves, shares, schedules, exports, or changes anything.

It will not:
• Make academic decisions for you.
• Register you for classes.
```

- Send messages without your review.
- Share your private information without permission.

Controls:

- AI enabled / disabled where applicable.
- Selected sources only / approved course sources / institution sources options.
- AI history retention and deletion control.
- Optional memory categories.
- Report inaccurate or unsafe output.

7.4 First-week experience

Day	Semester moment
Day 0	First clear action and saved plan/context
Day 1	Today briefing and one useful prompt
Day 2	Ask student to review sources or add one deadline/course
Day 3	Suggest an advisor agenda, study plan, or campus resource if relevant
Day 5	Gentle check-in: “Did this help you understand what to do next?”
Day 7	Weekly reset: priorities, schedule, deadlines, study blocks, opportunities

Do not flood new users with notifications. The goal is confidence and practical progress—not engagement for its own sake.

8. AI ROI scorecards

Student-facing ROI

Workflow	Leading metric	Outcome metric	Guardrail metric
Registration Copilot	Plan drafts created	Conflicts resolved / backups saved	Incorrect-source reports
Study Studio	Study assets created	Study sessions completed / helpfulness	Citation coverage / integrity blocks
Advisor Prep	Agendas created	Advisor usefulness / follow-up completion	Oversharing incidents
Campus Navigator	Search-to-action	Successful resource handoff	Stale-content reports
Career Evidence	Skills confirmed	Portfolio/application milestone	Invented-claim corrections

Internal team ROI

Workflow	Leading metric	Outcome metric	Guardrail metric
Engineering Triage	Incidents summarized	MTTR reduction / recurrence reduction	Secret or PII exposure incidents
RFP Copilot	Draft responses	Response cycle time / conversion	Unsupported-claim corrections
Content Governance	Stale items identified	Freshness improvement	Incorrect auto-draft rate
Talent Operations	Interview kits generated	Prep time reduction	Bias/quality issue reports

ROI calculation

$$\text{Workflow ROI} = \frac{\text{Quantified benefit} - \text{Direct workflow cost}}{\text{Direct workflow cost}}$$

Include direct cost of model inference, embeddings, retrieval, storage, moderation, support, human review, vendor fees, and engineering maintenance. Do not count raw usage as value; count completed, useful, safe outcomes.

9. Implementation roadmap

Phase 1 — Safe foundations

AI provider registry and vendor scorecards
Workflow cards and risk classification
Central policy engine
Source access and retrieval controls
Citation/source-card components
AI audit event schema
Feature flags and kill switches
Feedback/report-issue workflow
Prompt-injection and tool-use test suite

Phase 2 — Student value

Ask Semester global search
Syllabus/material ingestion with review
Study assets and source-linked explanations
Advisor agenda drafting
Registration and Path explanations
Campus resource navigation
Career evidence drafting

Phase 3 — Controlled actions

AI Action Composer
Study block drafts
Calendar-event drafts
Portfolio and application drafts
Advisor-share drafts
Exact confirmation UI and audit binding

Phase 4 — Institution governance

Tenant AI policies
Faculty course policies
Approved source packs
Model routing configuration
AI quality/evaluation console
Usage and cost dashboards
Trust Center AI documentation
Institution-specific controls and retention settings

10. Definition of done

An AI workflow is ready for production only when:

- [] A real user job and trigger are defined
- [] The non-AI workflow is usable
- [] Sources and data classification are defined
- [] Authorization happens before retrieval
- [] Input/output schemas are validated
- [] Citations or limitation notices are implemented
- [] Academic-integrity rules are enforced where relevant
- [] Sensitive/restricted data is excluded or specially approved
- [] Human confirmation exists for consequential action
- [] Prompt-injection and misuse tests pass
- [] Feature flag and kill switch exist
- [] Audit events and user feedback exist
- [] Accessibility acceptance criteria pass
- [] Cost, latency, quality, and safety metrics are monitored
- [] A named product, security, and operational owner approves launch

11. Final principle

The strongest AI strategy for Semester is not autonomous behavior. It is **earned trust**.

Semester Intelligence should consistently turn a confusing university signal into a sourced, explainable, student-controlled next step—while keeping people, institutions, policies, and safety boundaries in control.

